

Nitish Nagesh

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EDUCATION

Ph.D. Computer Science, University of California Irvine, GPA: 3.90/4.0	Sep 2021 - Dec 2026
M.S. Computer Science, University of California Irvine, GPA: 3.83/4.0	Sep 2021 - June 2023
M.S. Computer Engineering, Technical University of Munich, Germany GPA: 3.73/4.0	Sep 2018 - Dec 2020
B.E. Electrical Engineering, R.V. College of Engineering, India, GPA: 3.65/4.0	Sep 2012 - June 2016

PROGRAMMING SKILLS AND TOOLS

- Python, SQL, Pandas, Tensorflow, SciPy, Snowflake, Git, AWS, Confluence, Jira, Figma, Canva, Trello

PROFESSIONAL EXPERIENCE

Graduate Research Scientist, UC Irvine Institute for Future Health, Irvine, CA June 2022 - Present

- Build novel LLM-augmented causally fair synthetic data generation pipeline using Python, R to reduce bias in real-world health datasets; first-author publication in IEEE Medicine and Biology conference.
- Assisted in conceptualizing an LLM-powered framework for a diet recommendation chatbot using health data, achieving 92% effectiveness; co-authored publication in Elsevier Smart Health Journal.
- Integrated dietary guidelines and nutrition calculation tools into LLM-based health agents, enabling explainable diet risk assessments; co-authored publication in IEEE Medicine and Biology conference.
- Developed a causal inference framework using Python for a 3-year N-of-1 observational dataset to analyze the effect of caffeine on heart rate variability; first-author paper in computing conference.
- Implemented an image segmentation pipeline for a retina dataset using deep learning frameworks such as TensorFlow, Keras, SciPy achieving 31% accuracy for optic disk and 8% for fovea segmentation.

Research Data Scientist Intern, ResMed, San Diego, CA June 2024 - September 2024

- Built LLM-powered topic modeling framework to analyze quantitative data for 6,000 sleep apnea patients using LLMs and Langchain in Python, improving therapy personalization.
- Collaborated with an interdisciplinary team of 5 researchers to refine product claims related to pressure and wakefulness in sleep apnea patients; first-author abstract in American Thoracic Society 2025.

CEO & Founder, FoodMoodyAlly, Irvine, CA January 2023 - December 2023

- Lead planning, coordination and development of AI-driven food recommendation systems to improve mood of pregnant women & new mothers leveraging Python, Xcode, Trello, Git & AWS.
- Created user personas in Figma based on 30+ customer interviews using NSF I-Corps grant (\$5,000).
- Winner Product Design Competition (1st out of 17 teams), Mental Health Hackathon \$3,500 (1 of 30).

Data Scientist Intern, iHealth Labs, Sunnyvale, CA June 2023 - September 2023

- Led de-identification, aggregation, and analysis of data for 12,000 patients in a remote patient monitoring program, adhering to HIPAA guidelines, resulting in an academia-industry partnership.
- Collaborated with a cross-functional team to develop personalized interventions for diabetes and hypertension management; co-authored abstract and poster at American Heart Association 2024.

Software Engineer, Qualcomm, Austin, TX March 2021 - August 2021

- Developed a Python tool to parse 5,000 logs from the Qualcomm AI accelerator, reducing cycle time by 3x and lowering upstream production costs.
- Triaged and debugged failures in three ML accelerator SDKs through feature engineering and model evaluation, resulting in a 10% performance improvement across internal benchmarks.

Visiting Researcher, University of California San Diego, San Diego, CA September 2019 - February 2021

- Developed novel reliability-aware task allocation strategies for IoT networks using Python to reduce overall maintenance cost.
- Built a real-world IoT mesh network communicating via MQTT and Wi-Fi to measure impact of resource constraints on reliability leading to 3 co-authored publications in top-tier system conferences.

RELEVANT PUBLICATIONS

- [Submitted to AAAI 26] Nitish Nagesh*, Shakibhamedan* et al. "FairTabGen: Unifying Counterfactual and Causal Fairness in Synthetic Tabular Data Generation." * equal contribution
- Nitish Nagesh et al. "FairCauseSyn: Towards Causally Fair LLM-augmented Synthetic Data Generation.", IEEE Medicine and Biology Conference, Copenhagen, 2025.
- [Abstract] Nitish Nagesh et al. "Assessment Of AI/ML Approaches For Qualitative Analysis In Obstructive Sleep Apnea", American Thoracic Society, San Francisco, 2025.
- [Abstract] Yong Huang, Ajan Subramanian, Nitish Nagesh et al. "Patient Adherence to Self-Monitoring Practices and Glycemic Control - Findings from a Multiyear Digital Coaching Program", American Diabetes Association, 2025.
- [Under Review] Ajan Subramanian, Nitish Nagesh et al. "Remote Blood-Pressure Monitoring Improves Control in Stage II Hypertension", Journal of the American Heart Association, 2025
- Mahyar Abbasian, Nitish Nagesh et al. "Knowledge-Infused LLM-Powered Conversational Health Agent: A Case Study for Diabetes Patients", IEEE Medicine and Biology Conference, 2024.
- Zhongqi Yang, Nitish Nagesh et al. "ChatDiet: Empowering personalized nutrition-oriented food recommender chatbots through an LLM-augmented framework.", Elsevier Smart Health, 2024.
- [Abstract] Ajan Subramanian, Nitish Nagesh et al. "Long Term Remote Patient Monitoring Reduces Blood Pressure in Patients with Stage II Hypertension.", American Heart Association, 2024.
- Nitish Nagesh, Iman Azimi et al. "Towards Building Deep Personal Lifestyle Models using Multimodal N-of-1 Data.", 29th International Conference on Multimedia Modeling, Norway, 2023
- Ali Rostami, Nitish Nagesh, et al. "World food atlas for food navigation." Proceedings of the 7th International Workshop on Multimedia Assisted Dietary Management, Portugal, 2022.

LEADERSHIP AND SERVICE

Teaching Assistant, UC Irvine

Sep 2021 - Present

- Mentored 600+ students across writing, programming, data structures, and language courses, supporting instruction in collaboration with 5+ faculty.
- Designed and facilitated coding, design, and documentation support across multiple quarters, earning >90% student satisfaction in engagement and clarity.

Lead Mentor, Artificial Intelligence Club, UC Irvine

Sep 2021 - Present

- Initiated "How to apply to Grad School" series for 75+ students and reviewed 15+ personal statements leading to 5+ admission offers from CMU, UC Berkeley, etc.

DEI Representative and Peer Mentor, CS Department, UC Irvine

Sep 2021 - Present

- Ally for women in tech (GHC '23 scholar), peer mentor for 5 underrepresented students, led panel of 50+ attendees on ethical and societal challenges in computing.

Volunteer at Non-Profit, SMVA Trust, India

July 2016 - Aug 2018

- Led the "Feeding the Hungry" project delivering food and essentials to 10+ orphanages and senior centers to alleviate poverty and hunger.

HONORS AND AWARDS

- **UC Irvine** - Entrepreneur Award in Computer Science \$5,000 (1 per year), Fellowship to commercialize research \$2,500 (1 per year), Fellowship to coach student entrepreneurs \$5,000
- **TU Munich, Germany** - International Student Scholarship US\$2,500 (1 of 200) for academic excellence, Masters Thesis Scholarship US\$2,000 (1 of 50), Young Academy Scholar Cohort (1 of 40)
- **R V College, India** - Best Outgoing Student (1st in 70) for all round excellence, Academic Excellence Award (2nd in 70) for highest overall GPA, Most Innovative Thesis Award (2nd in 70)

INVITED TALKS

- LLMs/AI tools on a Daily Basis, Pacific Club, Feb. 2025, Newport Beach, CA
- Building Diet and Well-being Recommendation Systems for Lifestyle Management
 - Stanford University, Oct. 2023; IIIT-Delhi, India, Dec. 2022; INRAE, Paris, Oct. 2022; University of Barcelona, Spain, Oct. 2022